

Started on	Monday, 29 November 2021, 1:09 PM
State	Finished
Completed on	Monday, 29 November 2021, 1:09 PM
Time taken	20 secs
Marks	0/12
Grade	0 out of 100

Question 1

Not answered

Marked out of 1

A psychology trainee administers Beck's depression inventory to detect unidentified depression among the general population. She enrolls subjects for her study by interviewing people who attend the local sports centre and gymnasium. Which of the following bias is most likely?

Select one:

- ☐ Diagnostic suspicion bias
- ☐ Lead time bias
- ☐ Berkson's bias
- ☐ Measurement bias
- ☐ Selection bias

Your answer is incorrect.

It is likely that people who have an active lifestyle by attending leisure centres and gymnasium differ from the general population that has a mixture of active and inactive people. Studying unidentified depression by using a select sample as this will lead to selection bias.

The correct answer is: Selection bias

Question 2

Not answered

Marked out of 1

Two-stage screening surveys are commonly used to estimate the prevalence of psychiatric disorders in the community. In the first stage, questionnaires are sent to a large sample of subjects. The results from this stage are scored and used to identify probable cases who are then clinically interviewed. Which of the following is most likely to obscure true prevalence rates in such studies?

Select one:

- ☐ Recall bias
- ☐ Berkson's bias
- ☐ Non-response bias
- ☐ Confounding bias
- ☐ Diagnostic suspicion bias

Your answer is incorrect.

Selection bias occurs when study participants are selected in such a way that the association between exposure and disease differs for those who participate and those who do not participate in the study. One type of selection bias seen especially in postal surveys is non-response bias where responders who send their answers back to the researcher systematically differ from non-responders. It is well established that for lifestyle and behaviour related issues, non-responders often tend to have more problems than responders. This applies to mental health issues as well.

The correct answer is: Non-response bias

Question 3

Not answered

Marked out of 1

A regional affective disorders unit decides to find out patients' satisfaction with their service. They send out an easy-to-use one-page questionnaire to all service users on their list. A specialist trainee collates the results to present as a poster. She reports 85% satisfaction of service users with a questionnaire return rate of 65%. Which of the following is correct?

Select one:

- ☐ The result is most likely to be an under-estimation of the satisfaction
- ☐ Satisfaction is a normally distributed variable
- ☐ Random selection of patients to post questionnaires will be more representative
- ☐ Systematic differences exist between responders and non-responders
- ☐ Reminders do not increase response rates of postal surveys

Your answer is incorrect.

Satisfaction is not a normally distributed variable. It is likely that the 'most satisfied' group responded while least satisfied group abstained from posting the questionnaire back. This non-response bias could twist the results with an overestimation of actual satisfaction rates. Sending reminders can increase response rates. But randomly selecting responses that are returned will not provide any respite from non-response bias as the pool from which random selection takes place is in itself biased (i.e. they are all from 'most satisfied people'). Similarly randomly selecting people to send the questionnaire will also be of no use as still the response rate may be low, with non-satisfied people abstaining from returning the questionnaire. www.biomedcentral.com/1472-6963/4/31

The correct answer is: Systematic differences exist between responders and non-responders

Question 4

Not answered

Marked out of 1

A researcher's expectations regarding treatment effect may result in differential interactions with some subjects leading to a change in subject's response in favour of the research hypothesis. This is known as

Select one:

- ☐ Hawthorne effect
- ☐ Social desirability bias
- ☐ Halo effect
- ☐ Pygmalion effect
- ☐ Placebo effect

Your answer is incorrect.

The Pygmalion effect is also called the experimenter expectancy effect. Pygmalion is Bernard Shaw's play where a professor bets and succeeds in teaching a poor flower girl to speak and act like a higher-class lady. Mere awareness of the experimenter's expectations regarding treatment effect may result in a change in subject behavior in the direction of experimenter hypothesis. Halo effect refers to a type of primacy effect in which first impressions or answers influence subsequent answers in a questionnaire or interview. Gray GE. Evidence-based psychiatry. American Psychiatric Publishing, 2004. Pg 57

The correct answer is: Pygmalion effect

Question 5

Not answered

Marked out of 1

A clinician involved in a trial could provide additional care to patients in one treatment group but not to those in the other treatment group. This effect is called cointervention bias. This can be reduced by

Select one:

- ☐ Using multiple placebo arms
- ☐ Randomisation
- ☐ Double blinding
- ☐ Allocation concealment
- ☐ Pragmatic trial methods

Your answer is incorrect.

A clinician involved in a trial may provide additional care (e.g., time, support, etc.) to patients in one treatment group but not to those in the other treatment group. This effect is known as cointervention or performance bias (Altman et al. 2001). In clinical trials, this effect (and also Pygmalion effect) can be minimized by blinding clinicians to the treatment being provided. Gray GE. Evidence-based psychiatry. American Psychiatric Publishing, 2004. Pg 57

The correct answer is: Double blinding

Question 6

Not answered

Marked out of 1

Both smoking and alcohol consumption are associated with senile cognitive impairment. A study investigating the effect of smoking on dementia fails to stratify exposed group according to drinking index. Which of the following can threaten the validity of conclusions derived from this study?

Select one:

- ☐ Selection bias
- ☐ Sampling bias
- ☐ Confounding bias
- ☐ Ascertainment bias
- ☐ Incidence-prevalence bias

Your answer is incorrect.

A confounder is a factor that is associated both with the exposure of interest and the risk of the disease being studied. Thus, the observed association between the exposure of interest and disease outcome may be incorrect. Confounding causes an error in comparisons between groups if the groups are not stratified or adjusted.

The correct answer is: Confounding bias

Question 7

Not answered

Marked out of 1

Despite multiple reminders, a postal survey of heroin users to study disclosure to DVLA had a low response rate. What can be done to improve the interpretation of conclusions from this survey?

Select one:

- ☐ Rephrase the questionnaire and send the new version to nonresponders
- ☐ Increase sample size by recruiting another random group of users
- ☐ Report the proportion of DVLA-disclosed responders as the lower limit of expected percentage of heroine users who have disclosed to DVLA
- ☐ Ignore non responders and report on the available data
- ☐ Compare and report the baseline characteristics of responders and non-responders

Your answer is incorrect.

It is well established that survey nonresponders are probably different from responders. It is likely that most nonresponders were dissatisfied with the service. Recruiting more people will not simply increase the response rate, and non-responders will probably exist in equal numbers even in the second random group. The most one can do in this situation is to compare what is known about responders with non-responders. This might throw some light on reasons for nonresponse and help deducting what direction their response might have taken.

The correct answer is: Compare and report the baseline characteristics of responders and non-responders

Question 8

Not answered

Marked out of 1

Mere observation of behaviour can improve performance. In medical research, this is called

Select one:

- ☐ Measurement error
- ☐ Desirability bias
- ☐ Rosenthal effect
- ☐ Hawthorne effect
- ☐ Berkson effect

Your answer is incorrect.

Hawthorne effect refers to the phenomenon that people participating in a study ('under observation') may improve their performance or behaviour simply because of the attention they receive. This does not occur for physiological parameters such as blood pressure but seen in psychological factors such as motivation, productivity and most measures of self-report. The name derives from a factory called the Hawthorne Works where experiments to improve worker productivity were carried out in the Industrial period. Surprisingly most of the changes (non-specific) made to workers' conditions temporarily improved productivity. See BMJ 2009; 338:b1164

The correct answer is: Hawthorne effect

Question 9

Not answered

Marked out of 1

A scandalous tabloid report suggested that Asperger's syndrome may be more common in infants born to mothers who have irregular periods in the year immediately preceding the birth. A medical charity funds a study to investigate this report. 100 mothers of children with Asperger's syndrome were interviewed with respect to their menstrual history. 100 mothers of similar age and gender distributed normal children were also interviewed. Which of the following may invalidate results in this study?

Select one:

- ☐ Information bias
- ☐ Attrition bias
- ☐ Confounding bias
- ☐ Selection bias
- ☐ Diagnostic suspicion bias

Your answer is incorrect.

Recall bias is a type of information bias. It can be due to rumination bias - the presence of disease influences the perception of its causes or exposure suspicion bias - the search for exposure to the putative cause once being diagnosed. Though it is classically associated with case-control studies, it can also occur in cohort studies (for example, workers who know their exposure to hazardous substances may show a trend to report more the effects related to them), and trials without participants' blinding. Delgado-Rodriguez and Llorca J. Bias. Journal Epid Comm Health 2004;58;635-641

The correct answer is: Information bias

Question 10

Not answered

Marked out of 1

Using five years follow-up data of 200 patients at a local eating disorder service, a researcher is able to trace up to 80% of subjects registered at the service between 1999-2002. The 5-year mortality in this group is no different from an age, socioeconomic class and gender-matched group in same locality obtained from employment registers. She concludes that mortality rates associated with anorexia are overestimated in other studies. Which of the following invalidates such conclusion?

Select one:

- ☐ Low sample size
- ☐ Lack of data on physical health
- ☐ Attrition rate
- ☐ Lack of adequate controls
- ☐ Recall bias

Your answer is incorrect.

It is highly likely that the untraced subjects in this study are actually deceased. In mortality studies, attrition bias is a very important concern.

The correct answer is: Attrition rate

Question 11

Not answered

Marked out of 1

The percentage of group members having quit smoking at post-test was found much higher in a group having received a quit-smoking training program than in the control group in an RCT. Which of the following must be carefully checked while interpreting the study results?

Select one:

- ☐ Recall accuracy
- ☐ Regression towards the mean
- ☐ Diffusion of treatment effects
- ☐ Repeated testing effects
- ☐ Percentage completing the quit smoking program

Your answer is incorrect.

Recall bias is unlikely in this scenario of a controlled trial; as the measurement pertains to percentage responders, regression towards the mean is also unlikely. But presence of high drop out rates can bias the results as it is likely that those who will not be successful in quitting will drop out of the program earlier than the ones who are more likely to succeed. This is important when considering translating evidence to clinical practice.

The correct answer is: Percentage completing the quit smoking program

Question 12

Not answered

Marked out of 1

It has been suggested that psychiatrists may investigate patients who use depot antipsychotics more often or more thoroughly for metabolic disturbances than patients who do not. If this is true, then an apparent association between depot antipsychotics and hypertriglyceridaemia can be due to

Select one:

- ☐ Recall bias
- ☐ Surveillance bias
- ☐ Interviewer bias
- ☐ Nonresponse bias
- ☐ Selection bias

Your answer is incorrect.

Surveillance bias, also called detection bias, occurs when one group is followed more closely than the other group. This could lead to an outcome being diagnosed more often in the more closely followed group, but not because it truly occurred more often in that group.
<http://ocw.tufts.edu/Content/1/lecturenotes/193317>

The correct answer is: Surveillance bias



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